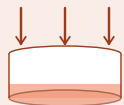


Extractive dilemma



depletion

Overuse of a shared resource depletes it.

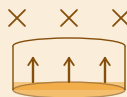
In humans

overfishing, overgrazing

In LLM agents

API budgets, balancing shared compute, tool calls, context

Contributive dilemma



no upkeep

Beneficiaries fail to maintain community goods.

In humans

open-source upkeep, unpaid moderation efforts

In LLM agents

helping at a personal cost, cleanup, verifying, volunteering

Second-order dilemma



who watches whom?

Monitoring is itself costly to provide.

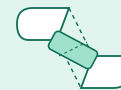
In humans

peer monitoring, costly security, expensive punishment

In LLM agents

auditing, sanctioning other agents, setting new norms

Repair dilemma



restoration

Who bears the cost of fixing harm?

In humans

restitution, policy reform
disaster recovery

In LLM agents

cleaning corrupted memory, retractions, reflect & undo changes, restoring what's lost